



Compact flash drives are USB-port devices that can hold dozens of megabytes and are small enough to fit on a key chain. The first flash drives required you to download drivers before you could plug them into the USB port. But the latest models offer true plug-and-play functionality, with no drivers necessary.

Battery research is focusing heavily on lithium chemistries, so much so that one could presume that all portable devices will be powered with lithium-ion batteries in the future. In many ways, lithium-ion is superior to nickel and lead-based chemistries and the applications for lithium-ion batteries are growing as

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Digital cameras are fast replacing their film-based cousins. Digicams have quickened the process of clicking and viewing images, and their costs have also come down substantially. This southward trend continues as newer and better cameras flood the market. All this has made the digital camera the instrument of choice for amateur photographers.

Though it's physically smaller than many other computer peripherals, the digital camera is not a simple piece of technology. From the lens and sensor processor, to the memory, and finally the storage device, everything works in sync. The final result is there for you to see on the LCD screen within a matter of seconds. Here, we go a little deeper into each of these components to see how their functions affect the overall quality of the camera—and that of your pictures.

The lens is the eye of any camera. It is what focuses the image and projects it onto the sensor, which in turn captures it. Image quality, therefore, depends on the quality of the lens.

Digital cameras come with either a fixed focus lens or a zoom lens. A fixed focus lens doesn't give you the option to zoom in on distant objects. A zoom lens, on the other hand, can zoom in on objects depending on the optical zoom capacity of the lens. Therefore, if you are planning to buy a camera for outdoor shots, especially landscape photography, it is advisable to opt for a model that has at least 5x optical zoom.

Nature photography enthusiasts need to pay attention to the macro focus range of the camera. A smaller macro distance will enable you to move closer to the object. Digital cameras also have a digital zoom option, which is nothing but software interpolation of an image. They do this by cropping the central part of the image and reducing its resolution, thus making it appear as if it's been zoomed into.